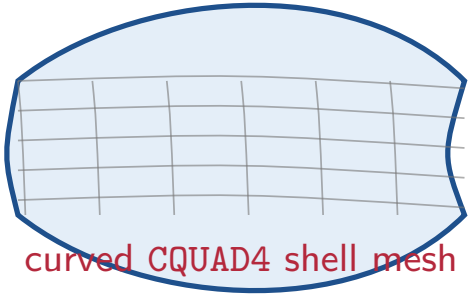
JFEM Shell-Element Formulation and Internal Gates

A Julia FEM solver with BDF-compatible front end

JFEM

Universidad de Sevilla

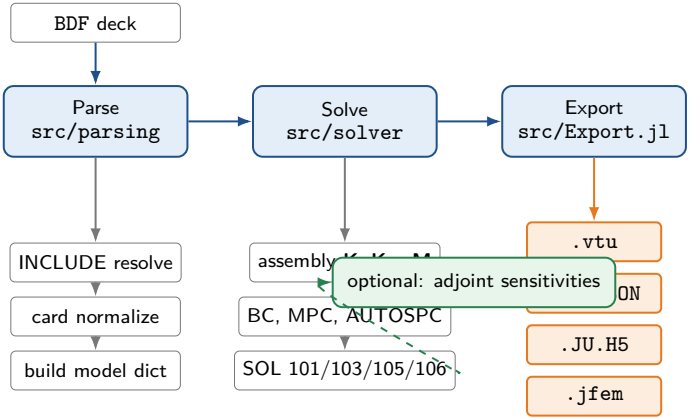
Updated June 12, 2026



Outline

1. JFEM at a glance — architecture
2. Solution sequences `SOL 101/103/105/106`
3. Shell kinematics, DOF layout, frames
4. Material handling: `MAT1/2/8`, `PCOMP` CLT
5. `PSHELL` field routing
6. Per-element `CQUAD4` dispatcher
7. MITC4 default kernel
8. φ_2 shear anti-locking
9. MacNeal RBF, MITC4+, MITC4-3D
10. DKQ, DKMQ, Adini, MIN4, hybrid stress, Hu–Washizu
11. `CTRIA3` macro-quad triangle
12. Higher-order `CTRIA6`/`CQUAD8` subdivision
13. Geometric stiffness $\mathbf{K}_G$ for `SOL 105`
14. Stress recovery and consistent mass
15. Gate catalogue
16. Validation status and open items

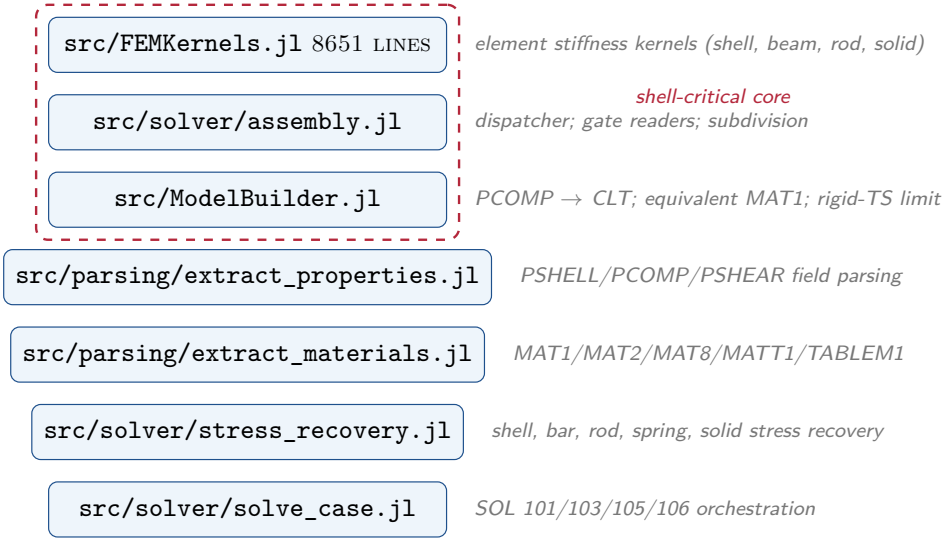
JFEM at a glance — three-stage workflow



Two design choices that drive everything

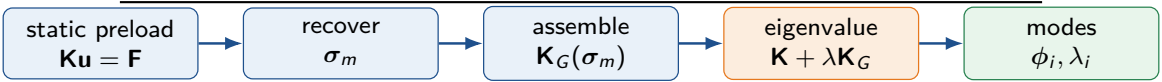
- The internal model is a JSON-serialisable nested `Dict` — the parse stage can be replaced by any external pre-processor.
- The dispatcher is purely per-element. Every shell element decides its own kernel from (`property`, `geometry`, `SOL`, `env`, `PARAM`).

Module map relevant to the shell layer



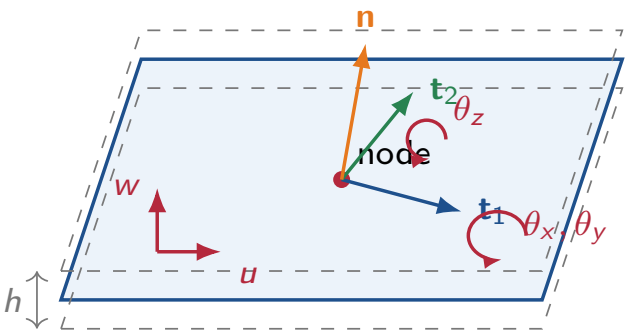
Solution sequences — equations and exports

SOL	Equation	Exports
101 static	$\mathbf{K}\mathbf{u} = \mathbf{F}$	displacements, reactions, $N/M/Q$, σ_{vM}
103 modal	$\mathbf{K}\phi = \omega^2\mathbf{M}\phi$	freqs, mode shapes, m_{eff}
105 buckling	$(\mathbf{K} + \lambda\mathbf{K}_G)\phi = \mathbf{0}$	buckling factors, mode shapes
106 nonlinear (exp.)	$\mathbf{K}_{\text{eff}}^{(k)}\Delta\mathbf{u} = -\mathbf{r}^{(k)}$	step history, residuals, line search



SOL 105 buckling path

Shell DOF layout (Reissner–Mindlin kinematics)



Displacement field:

$$\mathbf{u}(x, y, z) = \mathbf{u}_0(x, y) + z\,\boldsymbol{\theta}(x, y)$$

6 DOFs per node: $(u, v, w, \theta_x, \theta_y, \theta_z)$.

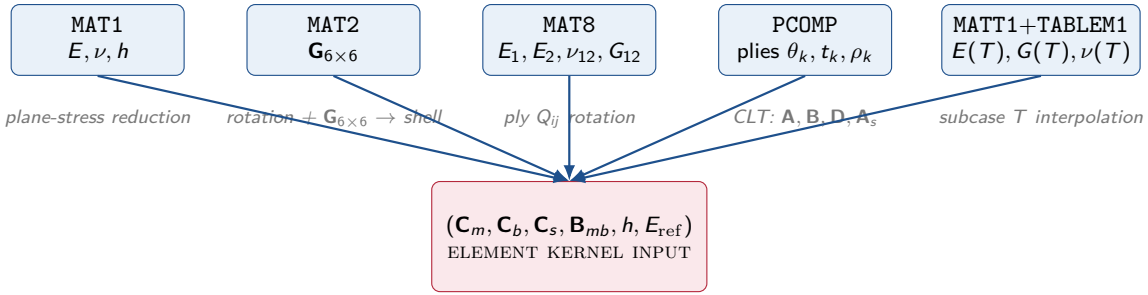
Transverse shear strains

$$\gamma_{xz} = \frac{\partial w}{\partial x} - \theta_y$$
$$\gamma_{yz} = \frac{\partial w}{\partial y} + \theta_x$$

Element sizes
`CQUAD4`: 24×24
`CTRIA3`: 18×18

θ_z is penalised through
$\alpha_{\text{drill}} = (\texttt{K6ROT}/10^5)\,Gh$

Material hierarchy reaching the shell kernel



Special case — MAT8 with blank G_{1Z}/G_{2Z} in PCOMP:
Reference convention $\Rightarrow$ rigid TS limit. JFEM default $\mathbf{C}_s = 2.5 \cdot \mathbf{C}_s^{\mathrm{Whitney}}$.
Override: JFEM_PCOMP_RIGID_TS_CS_SCALE or disable with JFEM_PCOMP_RIGID_TS_DISABLE=true.

Isotropic MAT1 constitutive blocks

For PSHELL/MAT1 with (E, ν, h) and PSHELL fields r_b=12I/T^3 (default 1), k_s=TS/T (default 5/6):

$$\mathbf{C}_m = \frac{Eh}{1-\nu^2}\begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{bmatrix}$$

$$\mathbf{C}_b = \frac{r_b\, Eh^3}{12(1-\nu^2)}\begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{bmatrix}$$

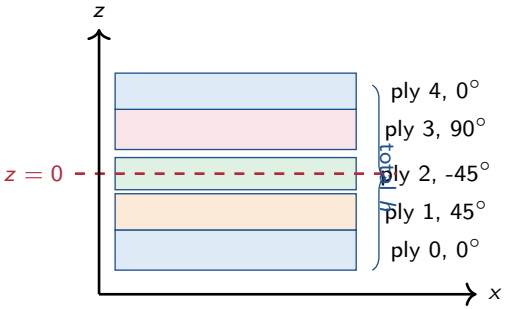
$$\mathbf{C}_s = k_s\, Gh\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad G = \frac{E}{2(1+\nu)}$$

Boundary cases

- MID2 blank $\Rightarrow \mathbf{C}_b = \mathbf{0}$, drill penalty also forced to 0 *(membrane-only shell)*
- MID1 blank $\Rightarrow \mathbf{C}_m = \mathbf{0}$ *(bending-only shell)*
- MID3 blank + MAT8 plies (PCOMP) $\Rightarrow$ rigid-TS override path

File evidence: FEMKernels.jl:1327-1335, extract_properties.jl:9-14.

PCOMP $\longrightarrow$ classical laminate theory



Stack reduction (per laminate)

$$\mathbf{A}_{ij} = \sum_k \bar{Q}_{ij}^{(k)} \, (z_k - z_{k-1})$$

$$\mathbf{B}_{ij} = \tfrac{1}{2} \sum_k \bar{Q}_{ij}^{(k)} \, (z_k^2 - z_{k-1}^2)$$

$$\mathbf{D}_{ij} = \tfrac{1}{3} \sum_k \bar{Q}_{ij}^{(k)} \, (z_k^3 - z_{k-1}^3)$$

$$\mathbf{A}_s = \sum_k \mathbf{G}_{\mathrm{sh}}^{(k)} \, (z_k - z_{k-1})$$

Stored on the property as: $\mathbf{C}_m$=**A**, $\mathbf{C}_b$=**D**, $\mathbf{B}_{mb}$=**B** *(or nothing if* $\|\mathbf{B}\| < 10^{-10}\|\mathbf{A}\|$*)*, $\mathbf{C}_s = k_s \mathbf{A}_s$.

Synthesised side data

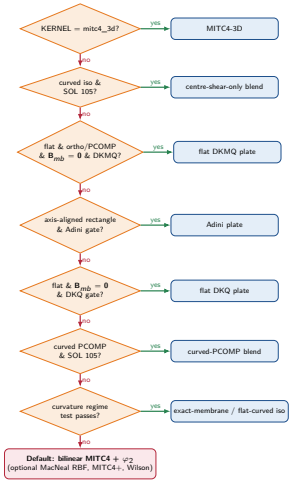
$E_{\mathrm{eq}} = A_{11}(1-\nu_{\mathrm{eq}}^2)/h$, $\quad \rho_{\mathrm{eff}} = \sum_k \rho^{(k)} t^{(k)}/h$, $\quad$ `PLY_DATA` retained for per-ply stress recovery.

PSHELL field routing

Field	Default	Role
MID1	—	membrane material id $\rightarrow \mathbf{C}_m$
T	—	thickness h (drives all thickness powers)
MID2	blank	bending material id; blank $\Rightarrow$ membrane-only
12I/T^3	1.0	bending ratio r_b on $\mathbf{C}_b$
MID3	blank	transverse-shear material; blank routes default
TS/T	5/6	transverse-shear factor k_s
NSM	0.0	non-structural mass per area $\rightarrow \mathbf{M}_{\mathrm{NSM}}$

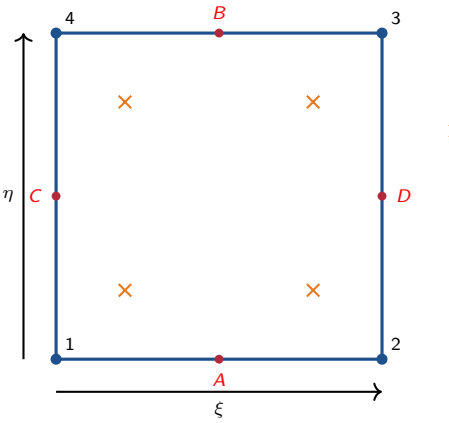
All field semantics are honoured directly from `extract_properties.jl:1-30`.

CQUAD4 dispatcher — decision tree



Each branch ANDs its gate with per-element geometry tests. Static-component scaling and MacNeal twist correction are layered *on top of* the selected kernel.

Default kernel: bilinear MITC4 with φ_2 anti-locking



Membrane $\mathbf{B}_m$: bilinear $\partial N_a/\partial x, \partial N_a/\partial y$ at 2×2 Gauss, optional Wilson modes $\phi_1{=}1{-}\xi^2$, $\phi_2{=}1{-}\eta^2$.

Bending $\mathbf{B}_b$: θ_x, θ_y at 2×2 Gauss; optional MacNeal twist correction $2\mathbf{B}_b^{\mathrm{gp}} - \mathbf{B}_b^{\mathrm{c}}$.

Shear $\mathbf{B}_s$: tied at A, B, C, D, interpolated to Gauss; covariant strains rotated: $\mathbf{C}_s^{\mathrm{phys}} = \varphi_2\,(J^{-1})^{\top}\mathbf{C}_s J^{-1}$

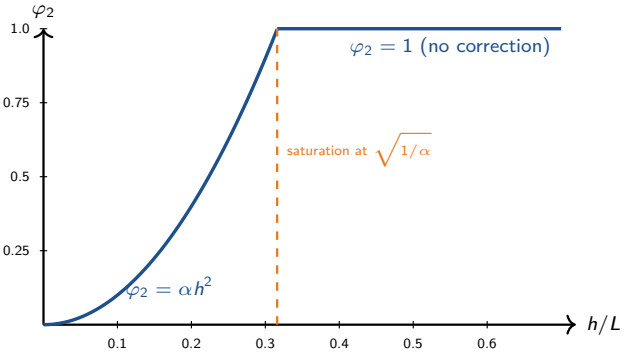
Anti-locking factor

$$\varphi_2(\xi,\eta) = \min\left(1,\ \alpha\,\frac{h^2}{4|\det J|}\right)$$

$\alpha =$ `PHI2_ALPHA[]` (default 10).

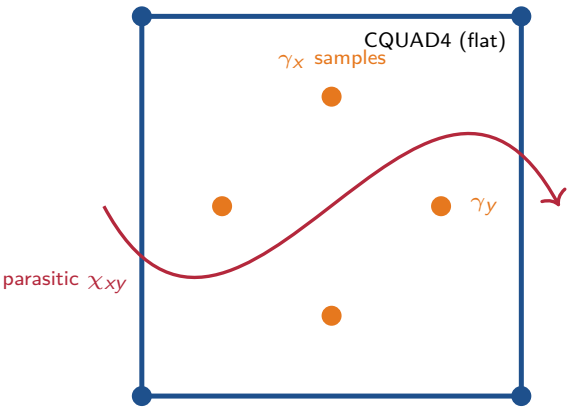
Drill: $\alpha_{\mathrm{drill}} = \frac{k_6}{10^5}\,G h$, $k_6 =$ `PARAM,K6ROT` (default 100).

φ_2 behaviour — thin vs. coarse meshes



- Thin shell, fine mesh: $h/L \ll 1 \Rightarrow \varphi_2$ drops, softening artificial shear stiffness $\rightarrow$ no shear locking.
- Thick / coarse mesh: φ_2=1, full Bathe–Dvorkin MITC4 is recovered.
- Saturation at $h/L = 1/\sqrt{\alpha} \approx 0.316$ with α=10.
- Setting α=0 disables the factor entirely (research mode).

Alternative: MacNeal residual-bending flexibility



Idea (MacNeal 1978): cancel parasitic-twist shear by augmenting $\mathbf{B}_s$ with a residual-bending flexibility block:

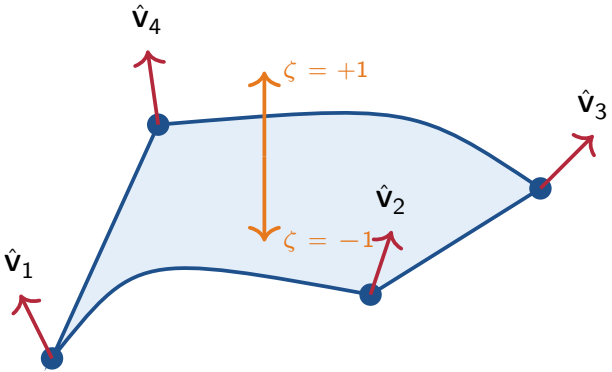
$$\tilde{\mathbf{B}}_s = \mathbf{B}_s^{\mathrm{MITC4}} + \frac{L^2}{12EI}\,\mathbf{1}$$

implemented as an explicit 4×4 matrix Z_b inverted in closed form.

Default tuning (2026-05-14)
`JFEM_Q4_MACNEAL_RBF_ZB_SCALE = 0.65`
Legacy 1.0 restored via `..._ZB_SCALE_LEGACY=true`.

Eligibility: warp $< 10^{-4}$ (relative), no PCOMP curvature. Bending-scale family of knobs (`_BENDING_SCALE`, `_ASPECT_SCALE`, `_CURVED_BENDING_SCALE`) calibrates per regime.

MITC4-3D — degenerate-shell kernel for curved geometry



curved 3D mid-surface

Position field (Dvorkin–Bathe 1984):

$$\mathbf{x}(\xi,\eta,\zeta) = \sum_a N_a\left(\mathbf{x}_a + \tfrac{h}{2}\zeta\,\hat{\mathbf{V}}_a\right)$$

Directors $\hat{\mathbf{V}}_a$: from `SNORM` or geometric corner-normal average.

Operator entry points (`FEMKernels.jl:658-900`):

- `stiffness_quad4_mitc4_3d_resultant_matrices` — laminate-resultant path
- `stiffness_quad4_mitc4_3d_ply_matrices` — per-ply integration (PCOMP)

Selectors: `..._SHEAR_MODE` (`covariant`/`physical`/`macneal`); per-block scales `..._{MEMBRANE,BENDING,SHEAR,DRILL}_SCALE`.

Status: *opt-in*; planned promotion to default to close curved-PCOMP gap.

MITC4+ assumed natural-strain membrane (Ko–Lee–Bathe 2017)

Standard MITC4 carries a parasitic $\xi\eta$ membrane term on *distorted* elements (e.g. tail-box trapezoids). MITC4+ replaces the membrane field with an assumed natural-coordinate strain:

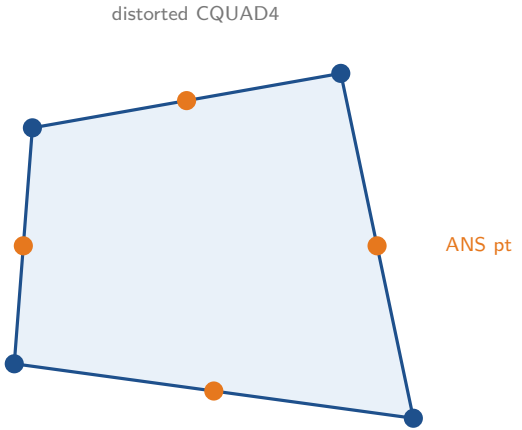
$$\varepsilon_{\alpha\beta}^{\mathrm{ANS}} = \sum_{k=1}^{4} N_k(\xi,\eta)\, \varepsilon_{\alpha\beta}^{(k)}$$

sampled at four ANS points, eliminating the locking mode.

Variants:

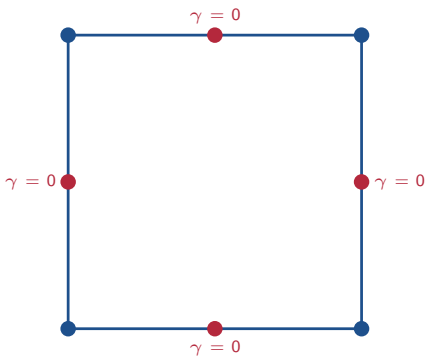
- `:mitc4plus_linear_J` — linear ANS using centre Jacobian
- `:mitc4plus_quadratic_J` — bilinear, J-dependent

Curved-shell variant gated by `JFEM_SOL105_EIG_MITC4PLUS_ALLOW_CURVED` (default `false`).



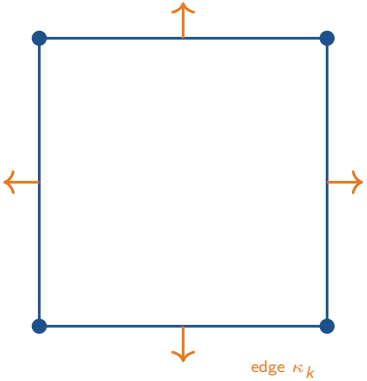
DKQ & DKMQ — discrete-Kirchhoff plate kernels

DKQ: $\gamma_{shear} = 0$ at edge midpoints



Thin-plate kernel. No transverse-shear deformability.
Active when flat & $\mathbf{B}_{mb} = \mathbf{0}$ & $\mathbf{C}_b \neq 0$.

DKMQ: per-edge hierarchical-rotation locking κ_k



$$\kappa_k = \frac{12}{L_k^2}\left[(\mathbf{H}_s^{-1})_{21}(\mathbf{H}_b)_{32} + (\mathbf{H}_s^{-1})_{22}(\mathbf{H}_b)_{22}\right]$$

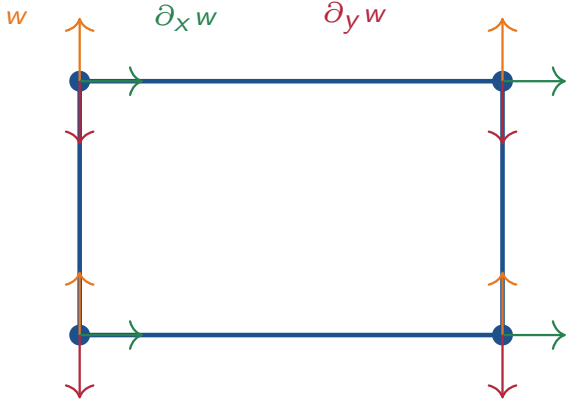
Includes shear; orthotropic/PCOMP-aware bending–twist coupling.
Default for flat orthotropic PCOMP.

Refs: Batoz/Bathe/Ho 1980, Katili–Maknun–Batoz–Ibrahimbegović 2018.

Adini rectangular plate & Tessler–Hughes MIN4

Adini (1961)

Axis-aligned rectangle only. 12-DOF bending: $(w, \partial_x w, \partial_y w)$ per corner. Full-cubic Hermite deflection, 4×4 Gauss–Legendre bending integration.



MIN4 (Tessler–Hughes 1983)

Anisoparametric: biquadratic w + bilinear (θ_x, θ_y); transverse-shear shape functions constrained to drop spurious mode.

Calibration knob

`JFEM_MIN4_CBMIN4 = 3.6`
(MYSTRAN reference)

Activation

`JFEM_Q4_KERNEL = min4`

Both kernels are **research switches**, available as opt-in alternatives to the MITC4 default.

Hybrid stress (Pian–Sumihara) & Hu–Washizu mixed

Hybrid stress (5 modes)

$$\mathbf{K}_e = \mathbf{G}^\top \mathbf{H}^{-1} \mathbf{G}$$

$$\mathbf{H} = \int P^\top S^{-1} P \, \mathrm{d}A, \quad \mathbf{G} = \int P^\top B \, \mathrm{d}A$$

Three variants in JFEM:

- `...membrane_hybrid_stress...`
- `...bending_hybrid_stress...`
- `...membrane_bending_hybrid_stress...`

Hu–Washizu (3-field)

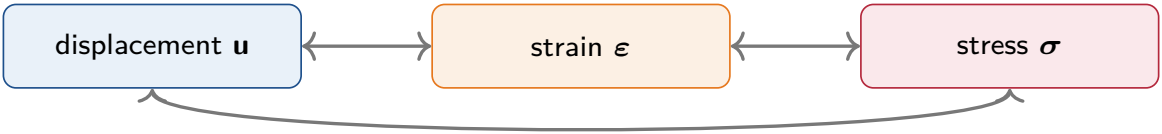
$$\Pi_{\mathrm{HW}}(\mathbf{u}, \boldsymbol{\varepsilon}, \boldsymbol{\sigma}) \;\rightarrow\; \text{stationary}$$

Independent fields:

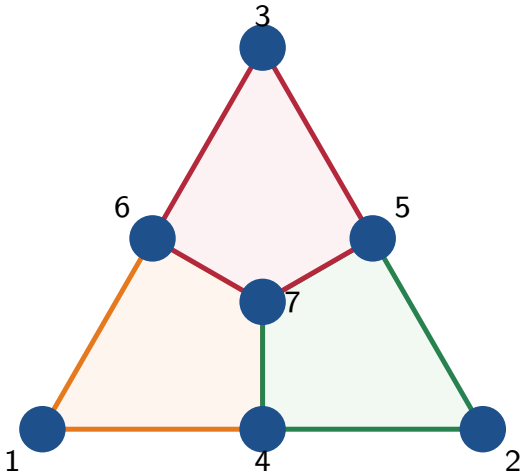
- displacement $\mathbf{u}$
- strain $\boldsymbol{\varepsilon}$ (5 modes)
- stress $\boldsymbol{\sigma}$ (5 modes)

Activation: `JFEM_Q4_KERNEL=huwashizu`.

Used as a *clean baseline* for cross-checking MITC4, MacNeal, MIN4.



`CTRIA3` as a macro-quad superposition



Algorithm (`FEMKernels.jl:6782-6862`):

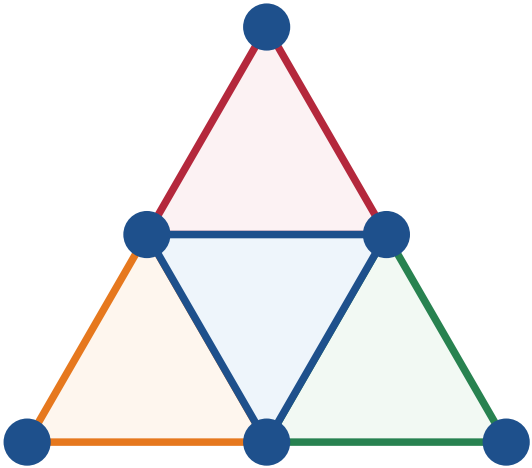
1. 7 virtual points: 3 corners + 3 edge mids + 1 centroid
2. 3 overlapping macro-quads:
   `TRIA3_MACRO_QUADS = ((1,4,7,6),(2,5,7,4),(3,6,7,5))`
3. Each macro-quad runs through `stiffness_quad4_matrices`
4. Keep plate DOFs only (w, θ_x, θ_y) per macro-quad
5. Assemble 21×21 macro stiffness
6. Condense the 12 internal DOFs $\rightarrow 9 \times 9$

$$\mathbf{K}_{\text{cond}} = \mathbf{K}_{ee} - \mathbf{K}_{ei}\mathbf{K}_{ii}^{-1}\mathbf{K}_{ie}$$

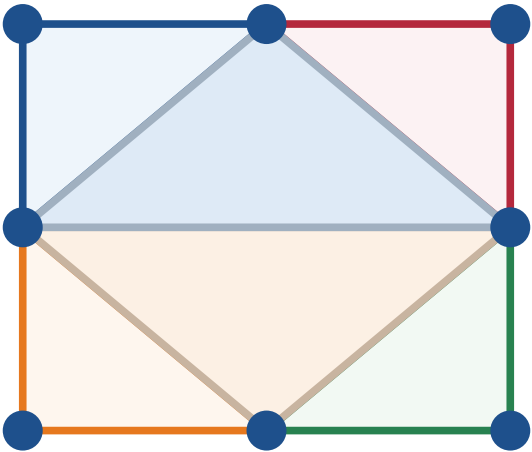
The condensation map $\mathbf{A}_{\text{int}} = -\mathbf{K}_{ii}^{-1}\mathbf{K}_{ie}$ is retained for pressure loads, stress recovery, and $\mathbf{K}_G$.

Higher-order shells via subdivision

`CTRIA6` → 4 `CTRIA3`

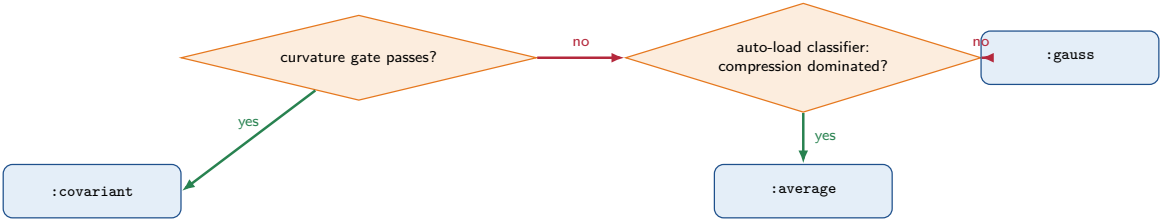


`CQUAD8` → 6 `CTRIA3`



Quadratic basis *not exploited*; effective element order = 1. Trade-off: input compatibility at modest implementation cost. A dedicated quadratic kernel is a candidate development item, not scheduled.

$\mathbf{K}_G$ stress-field classifier (`JFEM_KG_QUAD4_STRESS_FIELD_MODE`)



Tests in the auto-load classifier (all must pass for averaged σ_m):

- per-element geometry test;
- both principal averages in compression;
- biaxial compression;
- shear-resultant ratio below 2%;
- global load dominated by one axis with the matching sign.

$\mathbf{K}_G$ extras: `rot_grad_scale`, principal-TS coupling (`JFEM_KG_SHELL_TRANS_MODE=principal`), PCOMP N_{xy} scale.

$\mathbf{K}_G$ assembly recipe for shells

Per element, per Gauss point:

$$\mathbf{K}_G^{(g)} = \int_A (\partial N)^\top \begin{bmatrix} \sigma_{xx} & \sigma_{xy} \\ \sigma_{xy} & \sigma_{yy} \end{bmatrix} (\partial N)\, \mathrm{d}A$$

Per-mode contributions added on top:

translational
$(\partial N)^\top \boldsymbol{\sigma} (\partial N)$

rotational
rot-grad branch

principal TS
`...TRANS_MODE`

membrane-incomp
Wilson bubble

plate variants
(DKMQ, DKQ, Adini)

Stress recovery routes feeding $\mathbf{K}_G$

- `quad4_membrane_force_field` — 2×2 Gauss, optional Wilson condensation
- `quad4_membrane_force_field_triangle_recovery` — CST triangle averaging modes
- `quad4_membrane_force_field_covariant` — 3D coords + metric tensor
- `quad4_membrane_cst_resultant` — exact constant-strain triangle

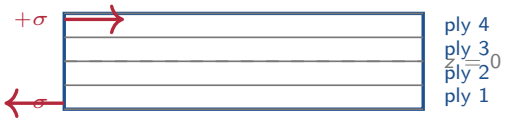
Shell stress recovery: what gets exported

Quantities recovered (`stress_recovery.jl:163-430`):

- membrane resultants (N_{xx}, N_{yy}, N_{xy})
- bending moments (M_{xx}, M_{yy}, M_{xy})
- transverse shear (Q_x, Q_y) at centre
- top/bottom surface $\sigma_{xx}, \sigma_{yy}, \sigma_{xy}, \sigma_{vM}$
- **Per-ply** stresses at top/bot of every ply (PCOMP), rotated back by θ_{ply}

Core helpers

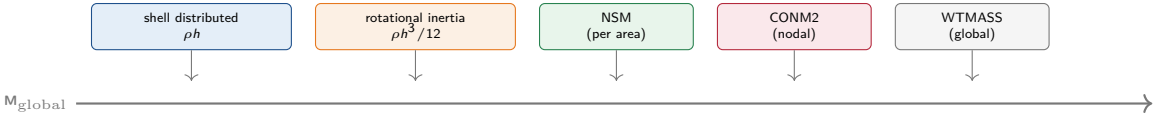
- `stress_strain_quad4` — full Q/T/B recovery
- `stress_strain_tria3` — via macro-quads + $\mathbf{A}_{\text{int}}$ map
- `quad4_mitc4_center_shear_resultant` — (Q_x, Q_y) at $\xi{=}\eta{=}0$



Resultant theory follows Simo–Fox–Rifai 1989: N, M, Q direct from constitutive blocks, not through-thickness re-integration.

Consistent mass for shells

- `consistent_mass_quad4` (`FEMKernels.jl:7857`): 2×2 Gauss; translational $\propto \rho h$, rotational $\propto \rho h^3/12$, drilling $= 0$.
- `consistent_mass_tria3` (`FEMKernels.jl:7903`): analytical on the CST; diagonal $\rho hA/6$, off-diagonal $\rho hA/12$.
- Non-structural mass from `PSHELL/NSM` or `PCOMP/NSM` adds to diagonal as $\mathrm{NSM} \cdot A$.
- `PARAM,WTMASS` multiplies the final $\mathbf{M}$.
- `CONM2` concentrated mass added separately.



Internal gates — categories at a glance

Globals
PHI2_ALPHA
MITC4PLUS_ALLOW_CURVED

Curvature
...CURVED_JACOBIAN
...CURVATURE_FILTER

MacNeal RBF
...ZB_SCALE = 0.65
...BENDING_SCALE_*

MITC4-3D
...SHEAR_MODE
...{M,B,S,D}_SCALE

Flat-iso
...MEMBRANE_INCOMP
...DKMQ

Flat-PCOMP
...PLATE_BRANCH
...EXACT_MEMBRANE

Drill / K6ROT
PARAM,K6ROT=100
...EIG_FLOOR=300

PCOMP rigid-TS
...DISABLE
...CS_SCALE=2.5

Marguerre
...COUPLING
...CONVENTION

Kg
...STRESS_FIELD_MODE
...AUTO_AVG_LOAD_*

Static-component
...{CM,CB,CS}_SCALE
...EID_FILTER

Hotspot
...NEIGHBOR_PID
...V2_MIN/MAX

$\approx$ 70 named env vars and PARAM entries, organised into 12 categories. Full catalogue in the companion paper (§12).

Example: switching default kernel & tuning rigid-TS

```
# Use MITC4-3D as the default kernel for SOL 105 eigenvalue assembly
export JFEM_Q4_KERNEL_EIG=mitc4_3d
export JFEM_Q4_MITC4_3D_PLY_INTEGRATION=true

# Override PCOMP rigid-TS uniform scale (default 2.5)
export JFEM_PCOMP_RIGID_TS_CS_SCALE=2.0

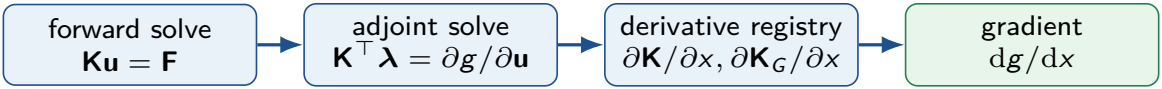
# Disable rigid-TS entirely (Whitney softening recovered)
export JFEM_PCOMP_RIGID_TS_DISABLE=true

# Localisation filter: scale Cs only on a named cluster of EIDs
export JFEM_Q4_STATIC_COMPONENT_EID_FILTER="51100,51101,51102"
export JFEM_Q4_STATIC_COMPONENT_CS_SCALE=2.5
export JFEM_Q4_STATIC_COMPONENT_NEIGHBOR_PID_PREFIXES="MODEL_A_,MODEL_B_"
```

Gates are read *once per element* at assembly time. Per-pass overrides (`..._STATIC`, `..._EIG`, `..._KG`) allow the static $\mathbf{K}$, eigen $\mathbf{K}$, and $\mathbf{K}_G$ to use different kernels in the same run.

Adjoint sensitivities — shell variables

Variable	Back-end	Interpretation
`shell_thickness`	`:ad_forward`	ForwardDiff on CTRIA3 + default CQUAD4
`material_E`	`:analytical`	Exact $\partial \mathbf{K}/\partial E$
`material_NU`	`:ad_forward`	ForwardDiff on standard shell; exact $\partial \mathbf{K}_G/\partial \nu$ on specialised branches
`pcomp_ply_thickness`	`:laminate_exact`	Differentiates $\mathbf{A}, \mathbf{B}, \mathbf{D}, \mathbf{A}_s$ directly
`pcomp_ply_angle`	`:laminate_exact`	Same path, with θ_{ply} AD
`node_coord`	`:full_model_fd`	Geometric perturbation + reassembly



Validation harness

Broad corpus (private validation regression tools):

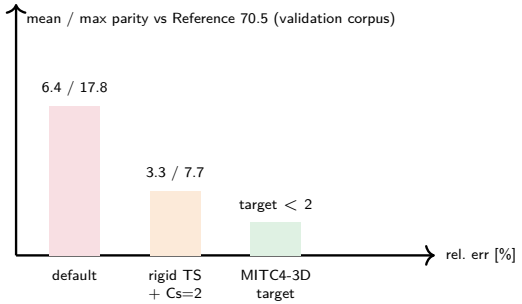
Category	Count	Interpretation
Total decks processed	3997	full Reference F06 paired set
OK (workflow + parity)	2048	$\approx$ 67.8% raw / 74.4% effective
Solver error	973	numerical/procedural failure
Unsupported sequence	976	outside current scope (SOL 200, etc.)
Effective fragment	267	standalone INCLUDE pieces

Focused regressions cover:

- shell-level $\partial \mathbf{K}_G/\partial h$, $\partial \mathbf{K}_G/\partial \nu$
- raw-JSON bridge parity (BDF $\rightarrow$ raw JSON $\rightarrow$ model $\rightarrow$ solve)
- per-element MAC-based curved-shell tracking
- SOL 106 readiness gate (`verify_sol106.jl`)

Open items relevant to shells

- **Curved-PCOMP coupling**: tail-box strain case MAC= 0 on subcase 511002. Membrane-recovery resultant ratio 0.4 on strong-curvature patches.
- **Planned remedy**: promote MITC4-3D to default (acceptance criteria in `MITC4_3D_SHELL_IMPLEMENTATION_PLAN.md`).
- **Higher-order shells**: `CTRIA6`/`CQUAD8` sub-divided, not natively quadratic.
- **SOL 106**: limited to flat coplanar `PSHELL` `CTRIA3`/`CQUAD4`.
- **Triangle drill**: inherits from macro-quad, no independent triangular formulation.



Key takeaways

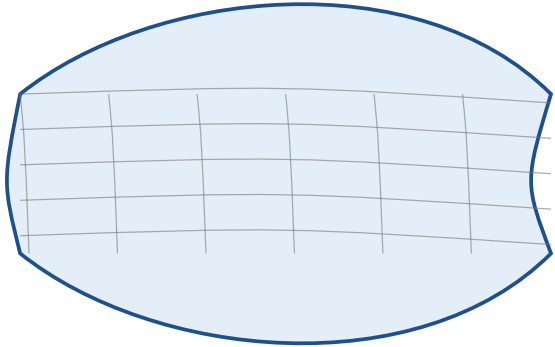
1. JFEM is a **three-stage Julia FE solver** with BDF-compatible BDF input and JSON-serialisable internal model.
2. Shell layer is **kernel-rich**: MITC4 (default) + MITC4+ + MITC4-3D + MacNeal RBF + DKQ + DKMQ + Adini + MIN4 + hybrid-stress + Hu–Washizu.
3. All shell properties — `MAT1`, `MAT2`, `MAT8`, `PCOMP` — reach the kernel as a uniform $(\mathbf{C}_m, \mathbf{C}_b, \mathbf{C}_s, \mathbf{B}_{mb})$ tuple plus per-ply data.
4. Kernel choice is a **per-element dispatcher decision**; ≈ 70 env vars + `PARAM`s tune it; defaults are conservative and validated.
5. `SOL 105` $\mathbf{K}_G$ has a **stress-field classifier** that picks Gauss / averaged / covariant recovery per element.
6. **CTRIA3** is built as a 3-macro-quad superposition with static condensation, sharing the entire `CQUAD4` kernel ladder.
7. Adjoint sensitivities: **mixed analytical + AD + FD**, with exact laminate-level paths for `PCOMP` ply variables.
8. Next big step: **MITC4-3D as default** for curved `PCOMP`, closing the largest remaining parity gap against Reference.

Selected references

- K.-J. Bathe, E. N. Dvorkin. *A four-node plate bending element based on Mindlin/Reissner plate theory and a mixed interpolation.* IJNME 21:367–383, 1985.
- E. N. Dvorkin, K.-J. Bathe. *A continuum mechanics based four-node shell element for general nonlinear analysis.* Eng. Comput. 1:77–88, 1984.
- K.-J. Bathe, A. Iosilevich, D. Chapelle. *An evaluation of the MITC shell elements.* Comp. Struct. 75:1–30, 2000.
- Y. Ko, P.-S. Lee, K.-J. Bathe. *A new 4-node MITC element for analysis of two-dimensional solids and its formulation in a shell element.* Comp. Struct. 192:34–49, 2017.
- R. H. MacNeal. *A simple quadrilateral shell element.* Comp. Struct. 8:175–183, 1978.
- T. J. R. Hughes, F. Brezzi. *On drilling degrees of freedom.* CMAME 72:105–121, 1989.
- J.-L. Batoz, K.-J. Bathe, L.-W. Ho. *A study of three-node triangular plate bending elements.* IJNME 15:1771–1812, 1980.
- Y. Lee, H.-M. Jeon, P.-S. Lee, K.-J. Bathe. *The modal behavior of the MITC3+ triangular shell element.* Comp. Struct. 153:148–164, 2015.
- I. Katili, I. J. Maknun, J.-L. Batoz, A. Ibrahimbegović. *Shear deformable shell element DKMQ24 for composite structures.* Comp. Struct. 202:182–200, 2018.
- A. Ibrahimbegović. *Stress resultant geometrically nonlinear shell theory with drilling rotations — Part I/II.* CMAME 118, 1994.
- J. C. Simo, D. D. Fox, M. S. Rifai. *On a stress resultant geometrically exact shell model. Part II.* CMAME 73:53–92, 1989.
- T. Belytschko, H. Stolarski, W. K. Liu, N. Carpenter, J. S.-J. Ong. *Stress projection for membrane and shear locking in shell finite elements.* CMAME 51:221–258, 1985.
- A. Tessler, T. J. R. Hughes. *An improved treatment of transverse shear in the Mindlin-type four-node quadrilateral element.* CMAME 39:311–335, 1983.
- J. N. Reddy. *Mechanics of Laminated Composite Plates and Shells*, 2nd ed., CRC Press, 2004.

Thank you

Questions?



Companion paper: JFEM/DOCS/Paper/ifem_2026_shell_paper.pdf